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Research article

A simulation-based framework for pre-installation solar PV feasibility assessment using long-term climatic data

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ABSTRACT

Generation of electricity from solar photovoltaic (PV) panels is an economical, clean, and renewable energy solution; however, achieving the desired power output during installation remains a challenging task. This study presents a method to estimate solar potential prior to the installation of a solar power plant at a given location. A discrepancy often exists between the installed and actual capacity of PV systems due to environmental factors, system losses, and PV cell efficiency. To address this issue, a 100 kW grid-connected solar PV system is modeled using an average Voltage Source Converter controller integrated with a Maximum Power Point Tracking algorithm based on the perturb and observe technique. Long-term climatological datasets of temperature and irradiance are obtained from the National Aeronautics and Space Administration (NASA) POWER database for the Bahawalpur region (Punjab, Pakistan). These data are used as input to the MATLAB/Simulink model to simulate power generation and power loss under varying conditions of direct normal irradiance, diffuse irradiance, and temperature. The results show good agreement with theoretical expectations. The generated power varies between 23.43% and 87.07%, while power loss ranges from 12.93% to 76.57% under different operating conditions. The proposed approach can be applied to other regions, subject to the availability of relevant climatic data.

1. Introduction

Energy generation trends are shifting due to the harmful effects of fossil fuel-based generation to renewable energies. The primary source of energy on Earth is the Sun. Therefore, all the natural cycles of the earth revolve around the sun in one or another way, leading from the process of photosynthesis, wind currents, carbonization, to the direct utilization of solar energy [1]. The importance of solar energy cannot be undermined, as it supplies an ample, clean source of energy without harmful greenhouse gas emissions. Also, it is a sustainable source of energy.

The global usage of solar electricity started back 3 decades ago, and it shows a significant boom from the year 1990, though the technology itself is half a century old. Also, in the field of solar energy and its efficient utilization, it has registered 86,000 patents up to the year 2017 [2]. This shows that scientists are keen to develop this technology to overcome the energy demand for the rest of the world in the future. There is ongoing research for the effective and efficient use of solar energy by the advent of the 21st century among researchers, which shows an improvement of materials and techniques in the infrastructure of solar plants for maximum power accumulation in the solar hours [3–6]. A key challenge is the portion of solar energy that is not converted into electricity from the solar photovoltaic (PV) power plant, which needs redressal for maximum capturing and utilization. This challenge is addressed by engineers, researchers, and scientists.

There is a potential gap between the research institutions and the engineering companies to solve the practical issues that relate to the solar industry, such as absorption of maximum solar radiation throughout the whole day, where the start and end of day may have maximum output of solar PV power plant, figuratively. Solar investors

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are less keen to invest in solar technology; they feel motivated to invest in wind technology for the high and timely investment returns. The high capital cost and large-scale investment are the only way to lower the price of the product for the access of a common man. Thus, solar technology remains less accessible in many regions [7].

The classical engineering practices in the field of solar energy are outdated, as they may not provide maximum extraction of solar irradiance for the high-power yields. It is limited adoption to find a solar monitoring system implanted by the assistance of Internet of Things and Programmable Logic Control, mounted Supervisory Control and Data Acquisition monitoring systems at residential and commercial levels [8]. The classics of switch gear technology are still used in most parts of the world, especially in the developing countries, where more than two-thirds of the population exists [9].

Generally, the PV solar power plants have a generic concern for having a significant difference between the installed and run capacities in terms of power generation. This power is lost in the process of generation, which needs to be addressed in detail to bring back the confidence among the investors for their pinpoint understanding of PV solar power plant technology [10–12]. The goal is to demonstrate a model that interprets and addresses the concerns of best engineering practices to provide a feasible solution for the maximum production of PV solar energy. Moreover, a model can be easily usable to adopt in the practical fields for the new erection of infrastructure or improving the existing PV solar plants. The main focus of this work is to find comprehensive framework to address and validate the power loss in the process of PV generation. This approach may support improved estimation of solar energy production. This study aims to quantify the difference between installed and actual power output. Research question is to find a hourly/daily time series solar potential whose results are coherent to the practicality and may be adopted for modern engineering practices without dubious frames.

The case study was performed in this research on the Quaid-e-Azam Solar Power Plant located in the Bahawalpur region. Initially, in the 1st phase of installation, this PV solar plant was erected to an installed capacity of 100 MW. The run capacity was observed to be as low as 18 MW, which also became the reason for the delay of the next phases to raise it to a number of 900 MW. Therefore, the government analyzed the mistake by installing this pilot project of 100 MW, and transferred it to private companies of Turkey and China [13,14]. Two reasons for the obstacle at the beginning of the next phase. Firstly, the solar survey work was inefficient enough for the site selection, as in summer regions, the average temperature of the day of Bahawalpur reaches 45 degrees centigrade, whereas 25 degrees is the nominal operation range for optimum efficiencies of PV solar arrays [15]. PV modules were designed for different climatic conditions, which could not satisfy the Bahawalpur environment at all [16]. Secondly, the government made all contract agreements to the installed capacity of the PV solar plant, whereas the run capacity was just 15–35% of the installed capacity [17]. This leads to a deficit in governmental finance expectations, and it led to the transfer of the Quaid-e-Azam Solar Power Plant to a public-private partnership. The main motive behind this research is to organize a step-by-step approach that assists governments and investors in improving confidence in PV technology, as it is the future of renewable energies.

In Hafez et al. [18], a solar feasibility of Saudi Arabia at 10 different locations to measure utility-scale with the assistance of software National Renewable Energy Laboratory. This way, they addressed the minimum cost of electricity by conducting software base solar feasibility. A hybrid model is conducted in Algeria for PV-Wind-Diesel-Battery Configuration on HOMER for ambient temperature, global solar radiation, and wind speed. This feasibility results in savings of 69% savings of fossil fuels [19]. The data logging review for prototype solar PV system feasibility shows the enhancement of solar absorption of 8 PV panels rated as 250 watts used at various sites [20–22]. RETScreen clean energy management software is used to analyze the performance of 50 MW solar PV system based on single-axis and dual-axis tracking. The Geographic Information System is accessed for data generation to provide resource mapping. However, the adopted algorithm is the Analytic Hierarchy Process and Multicriteria Decision Making. Whereas, the object of this study is to analyze the utility scale model feasibility for the reduction of fossil fuels by the enhancement of efficiency [23,24]. Whereas, this research highlights the problem of solar feasibility based on satellite data logging, which is missing in the literature reviewed. Also, the data set generated and used in MATLAB/SIMULINK briefs the power output for the solar's actual feasibility analysis. The research gap for this research is to prepare a feasibility study model that provides actual power loss and power generation of the solar PV power plant.

Existing studies on PV energy yield estimation mainly focus on system optimization, tracking mechanisms, or short-term irradiance inputs. While some models incorporate environmental and system parameters, they often rely on limited temporal resolution or site-specific conditions. The key gap addressed in this study is the use of long-term (multidecade) climatological data to develop a simplified simulation framework for energy yield estimation. This approach enables feasibility screening based on stable climatic trends, which is less explored in the current literature.

The scope of this study can be extended to any part of the world, prior to data extraction points and data validity for the specific region; for instance, in this study, the Bahawalpur region is the choice of selection.

1) The simulation based on long-term climatological data of a PV solar plant allows us to know the practical results for a specific region that can be extended to any part of the world.
2) This approach can support preliminary feasibility assessment prior to detailed field studies.
3) An approach is given to find a ‘home-based’ physical feasibility report for the solar potential anywhere in the world, without expending resources on physical surveys of the land for installing PV solar power plants.

This study highlights the gap between simulated power output and the actual generation of a PV solar power plant. This difference represents the portion of potential energy that is not realized in practice due to the combined effects of environmental variability, system losses, and non-ideal operating conditions. In standard PV system analysis, such discrepancies are commonly associated with loss mechanisms and performance ratios. For the Bahawalpur case study, the results demonstrate a clear difference between theoretical generation potential and observed plant performance.

2. Methodology

2.1. Generation of dataset

The foundation of this study lies in finding solar potential on an actual-time simulation. The predecessor work following this research uses the National Aeronautics and Space Administration (NASA) Power Data Access Viewer to narrate the idea of finding a solar potential [20,25]. Therefore, these studies partly lack actual-time results. This study depicts the idea to generate the results of PV solar power potential with respect to the coherent values of practicality, such as there may not be a significant difference between the practical values and simulation results produced by the assistance of software. Therefore, to achieve this arduous task, a 100 kW PV solar model is deployed that lays the basis for the findings of solar potential anywhere in the world. The user of this model must access the climatology data from any of the archives of solar research centers. The next step is to execute the data in the form of Excel sheets of 2-column formats, where each column represents the inputs of irradiance and temperature, respectively.

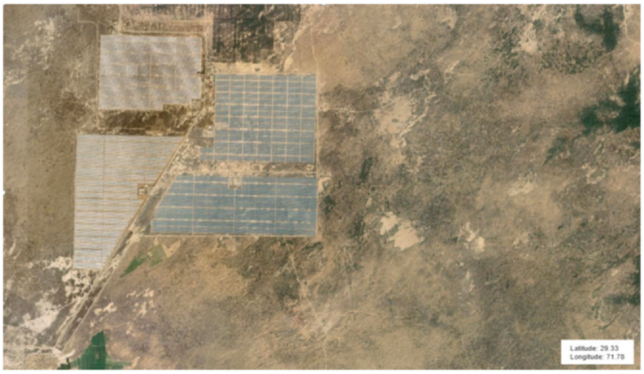

Fig. 1. Location (latitude and longitude) East longitude 71.78° and North latitude 29.33° for the extraction of climatology data from NASA. NASA = National Aeronautics and Space Administration.

Table 1
Conversion factor for solar radiation to solar irradiance

Conversion factor	W/m^2	$kW{\cdot}h/(m^2{\cdot}day)$	Sun hours/day
W/m^2	1	41.66666	41.66666
$kW{\cdot}h/(m^2{\cdot}day)$	0.024	1	1
sun hours/day	0.024	1	1

Fig. 1 shows the location coordinates of Quaid-e-Azam Solar Park under study. Data were obtained from the NASA POWER Data Access Viewer for the Quaid-e-Azam Solar Park, Bahawalpur (71.78° E, 29.33° N). An averaging scheme was applied to process the raw data based on the geographic coordinates of Bahawalpur extracted from the NASA POWER Data Access Viewer [22]. The climatology data of the previous 35 years was aggregated into 12 monthly data points representing months from January to December. The daily average of data for the different parameters gives a 30-point set for a month. Table 1 presents the conversion factors and conversion factor explicitly for radiation to irradiance of solar.

Averaging Scheme for Data Pre-Processing Steps Aggregate

$$Monthly\ Average_M = \frac{1}{24 * D_M * 35} \sum_{y=1}^{35} \sum_{d=1}^{D_M} \sum_{h=1}^{24} f(h,\ d,\ M,\ y)\# \quad (1)$$

Eq. (1) gives monthly average of 12 data points for 35 years data set; M = Month (January to December); Where; January = 1 to December = 12; D_M = number of days in month M = 31 (January, March, May, July, August, October, & December), M = 30 (February, April, June, September, & November), or M = 28 or 29 for February) and f (h, d, M, y) = hourly value for year y, month M, day d, & hour h.

The solar model works on a mechanism of MPPT-based perturb and observe (P&O) algorithm to convert a single-phase supply to a stable voltage profile of 3-phase with the aid of an average model of VSC that works on a theory of commutation. The boost converter nurtures the average voltage profile of single-phase to make the VSC system work at a stable rate. The low-voltage 3-phase supply is converted to a high-voltage supply through a substation that is further attached to a grid network for commercial uses. It may also act as a stand-alone system prior to the concept of micro-grid to meet the demand of the local market with the assistance of batteries to supply electric power in night mode. The PV solar plant model system used in this research supports the governments, investors, researchers, scientists, and engineers to find a solar electric potential without using any of the external resources. Thus, the testing cost is readily zero if anyone uses it for findings of the solar potential in any locality of the world. Moreover, a step-by-step approach is helpful for adopting the procedures mentioned in this research. The basic input parameters are the surface temperature and solar irradiance. However, these parameters can be subdivided into more than 2 categories to acquire the scientific results, which are given in Fig. 1. Nowhere, the maximum and minimum durations of sunshine are taken as 6.67 and 8.33 h per day, prior to the fact that 200–250 h of monthly sunshine is available in records of the Bahawalpur region [26].

Henceforth, the monthly average of the same month gives a dataset of 12 points. Furthermore, the rest of the climatology data for the years is converted into 12- point datasets by taking out averages. The purpose is to make it executable in a SIMULINK model of MATLAB to achieve desirable results. The raw data is modified using an average scheme to a 12-point dataset in 9 different parameters. Namely, 3 cases of direct normal irradiance (DNI) and Diffused Irradiance (DI) each that make up 6 parameters, and 3 cases of surface temperature of a PV module. Hence, these 9 parameters are organized and executed based on a turn-by-turn policy in a separate simulation. The refined data of 9 parameters is enlisted in Table 2. The organization and classification of these 9 parameters give the execution for conducting a simulation.

The climatology data for the previous 35 years is taken from the NASA Power Data Access Viewer of direct normal radiation (DNR), diffused radiation (DR), and the temperature of the Earth's surface. The solar radiation data can only be utilized in solar calculations if it is converted into solar irradiance. Therefore, the unit scheme mentioned in Table 2 is used for the conversion of radiation data into irradiance to use in the SIMULINK model of MATLAB software. It is to be noted that the unit for solar radiation is kWh / m^2 day, and the unit for solar irradiance is W / m^2.

2.2. Scheme to generate feasibility report

The main model is the 100-kW grid-connected PV array represented by an average model controlled by a voltage source converter (VSC) controller [27]. The maximum power point tracking (MPPT) technique

Table 2
Jan to Dec climatology data of power access

Months	Average diffused irradiance sunshine	Diffused irradiance sunshine (min)	Diffused irradiance sunshine (max)	Direct normal irradiance sunshine (min)	Direct normal irradiance sunshine (max)	Direct normal irradiance full day	Min. Temp. (°C)	Max. Temp. (°C)	Avg. Temp. (°C)
1	144	91.5	168	691.2	864	240	5.4	21.4	13.4
2	177	145.5	198	746.4	933	260	8.5	24.4	16.4
3	238.5	201	261	728.4	910.5	252	13.7	30.1	21.8
4	292.5	276	310.5	734.4	918	254	19.5	36.3	27.9
5	325.5	307.5	345	765.6	957	265	24.8	41.4	33.1
6	340.5	301.5	369	768	960	266	28.8	42.5	35.6
7	346.5	316.5	363	681.6	852	236	28.8	39.8	34.3
8	322.5	306	328.5	627.6	784.5	217	27.7	38.2	32.9
9	264	241.5	277.5	676.8	846	234	24.9	37.4	31.1
10	199.5	177	222	718.8	898.5	250	18.1	35.2	26.6
11	151.5	127.5	165	710.4	888	246	11.4	29.4	20.4
12	133.5	103.5	148.5	668.4	835.5	232	6.8	23.6	15.2

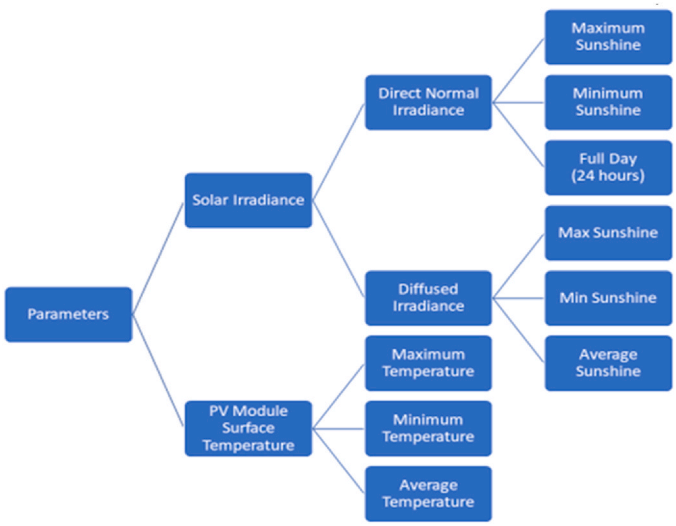

Fig. 2. Involved parameters for inputs to generate outputs.

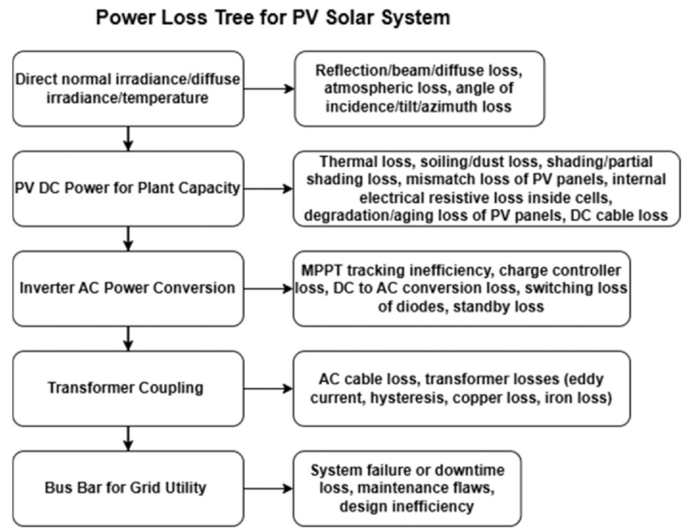

Fig. 3. Power loss tree for PV solar system. PV = photovoltaic.

is in consideration that work on the principle of a hill-climbing algorithm to track the maximum power with reference to the voltages. This technique is famous for the high percentage levels up to 98% to stream the Direct Current (DC) supply and provide the maximum power output at the boosting side [28,29]. The whole system works under the umbrella of 2 basic inputs: namely, the surface temperature of a PV module and the solar irradiance values of a specific region. In this model, the Bahawalpur region is chosen for experimentation to produce viable results. On the other hand, when the inputs are active and in a reasonable range, then the system produces electricity to the grid network. The P&O command helps to execute the Perturb & Observe algorithm for the work command window of a MATLAB [30].

The basic steps to cover this simulation are demonstrated in Fig. 2, which enlightens the sequential working of the model. The imported data from Power Access is treated as the input of temperature and irradiance for a 100-kW solar PV power plant. The output is the 3-phase power supplied to the local grid for distribution and utilization. The output power is measured at the grid-side Alternating Current (AC) bus of the inverter. A positive sign convention indicates power injected into the grid, while negative values correspond to reverse power flow during transient conditions. To ensure accurate analysis, only steady-state data are considered, and initial transient periods are excluded from statistical calculations.

2.3. Irradiance modeling approach

Plane of Array (POA) irradiation is the average accumulated irradiation from various physical weather stations located at the site plant over a specific period, normally a quarter of a year. This may be subject to the construction of irradiance, such as beam angles, diffusing factors, albedo, tilt/azimuth/incidence, plate erection angle, etc. [31]. Converting NASA variables to POA from 35 years of long-term data sets of DNI & DI may pass on many physical factors, which may change on a yearly basis, for the construction of irradiance POA. Whereas, DNI & DI may not specifically rely on the mentioned physicality factors, such as DI is less sensitive to tilt compared to direct irradiance, azimuth, and solar tracking. Also, DNI & DI are independent of the PV surface geometry, in contrast to POA. Moreover, DNI suggests an optimized pattern for the maximum limit of capturing solar rays [32]. This creates a sensitivity study of the solar PV system for considering the given inputs of DNI & DI. POA provides more physically representative results than DNI & DI for simulation. Averaging DNI & DI long-term data sets complies with simulative sensitivity analysis that relates to a featured approach towards finding the difference between install and run capacities of PV arrays. However, the assumptions of solar trackers are not included in this research to serve as a comparison analysis from DNI. Furthermore, sunshine is a narration used for the solar hours of the day.

Fig. 3 summarizes the power losses at every stage of the solar PV system; however, the input parameters for DNI, DI, and temperature are fixed for the sensitivity analysis in checking the power loss in this research, as per the model considered. Whereas other variables may be involved in measuring the power loss for future works.

3. Modeling, simulation, and system description

3.1. Main model

The main model is the 100-kW grid-connected PV array represented by an average model controlled by a VSC controller as shown in Fig. 4. The DC voltages collected by the PV arrays are boosted with the help of a boost converter. The MPPT technique is in consideration that work on a principle of hill climbing algorithm to track the maximum power with reference to the voltages. This technique is famous for the high percentage levels up to 98% to stream the DC supply and provide the maximum power output at the boosting side. The VSC provides the reference values to convert the boosted DC supply to a 3-phase AC supply, and then electricity is supplied to the nearby grid by stepping up with the help of a transformer. The whole system works under the umbrella of 2 basic inputs: namely, the surface temperature of a PV module and the solar irradiance values of a specific region. In this model, the Bahawalpur region is chosen for experimentation to produce viable results.

When the inputs are inadequate, the enabling command of MPPT is unable to register any changes that set the values to zero. On the other hand, when the inputs are active and in a reasonable range, then the system produces electricity to the grid network. The PandO command helps to execute the Perturb & Observe algorithm for the work command window of MATLAB.

The basic steps to cover this simulation are demonstrated in Fig. 5, which enlightens the sequential working of the model. Also, the Perturb & Observe algorithm used in the MPPT is given in the flowchart, given in Fig. 6 to understand the basic steps of an algorithm.

3.2. Boost converter

The boost converter helps to step up the DC supply collected from the PV arrays, and the modeling of a boost converter is illustrated in

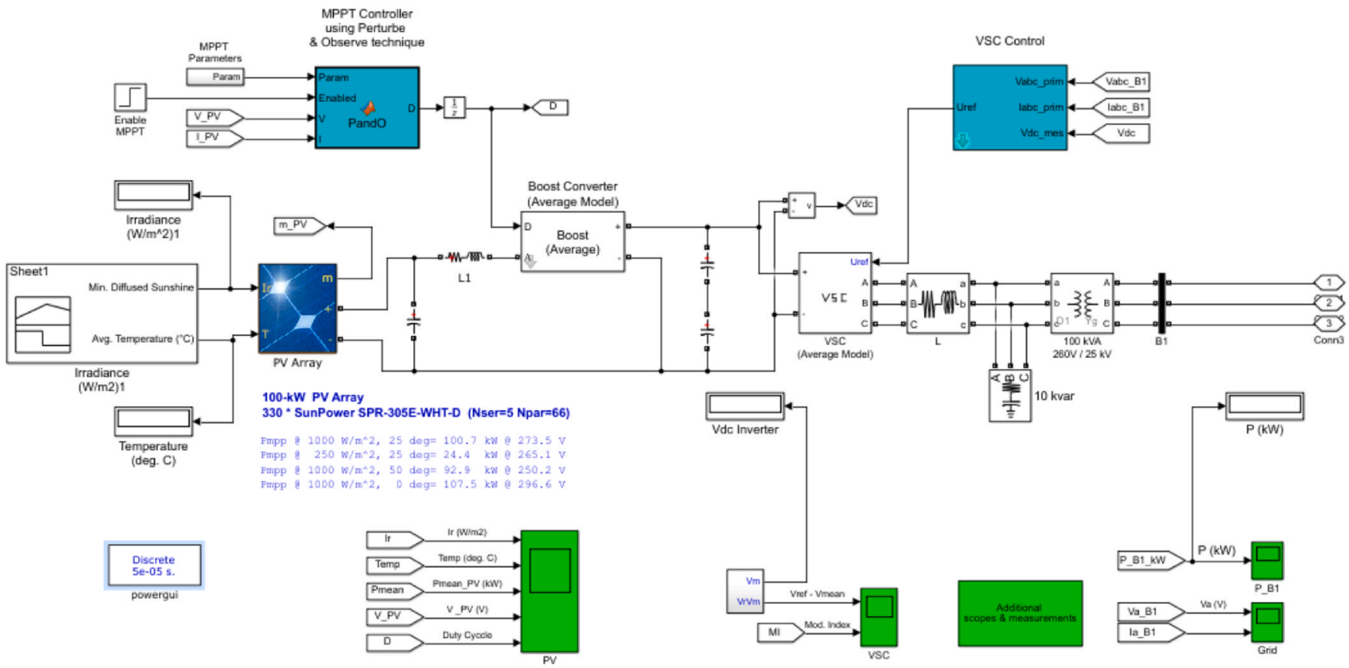

Fig. 4. Main model of a 100-kW grid-connected PV module [31]. PV = photovoltaic.

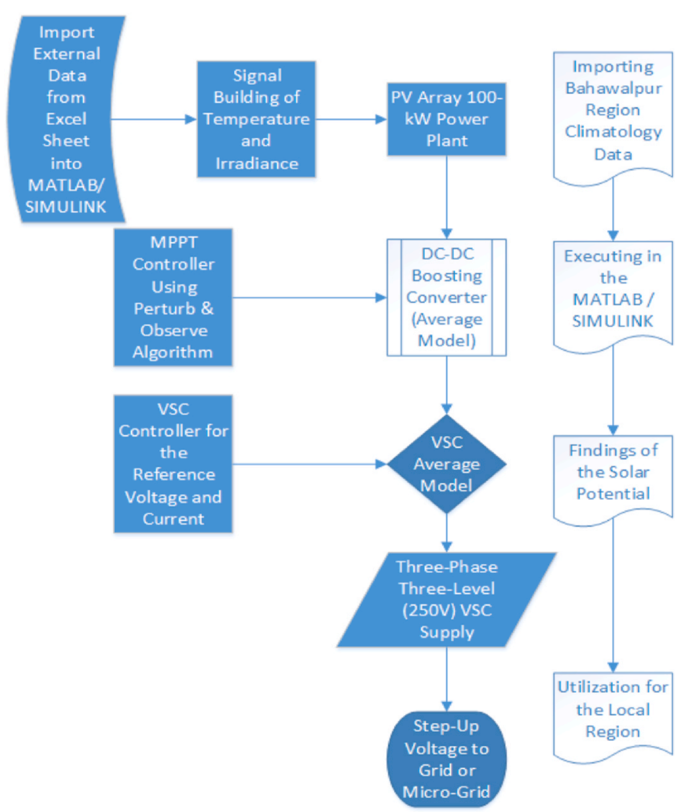

Fig. 5. Basic steps for the functionality of a solar PV model. MPPT = maximum power point tracking; PV = photovoltaic; VSC = voltage source converter.

Fig. 7. The scaling is applied with the unity constant values, and the gain is set to half the value to demonstrate the average values. The multiplying factors from the DC values of voltage and current give the electrical power, which is equivalent to the AC power supplied to the grid network.

3.3. VSC main controller

The VSC main controller helps the VSC to adjust the reference values with the phase-locked loop to convert the DC supply to the 3-phase AC supply. The current and voltage regulators register the smoothness to

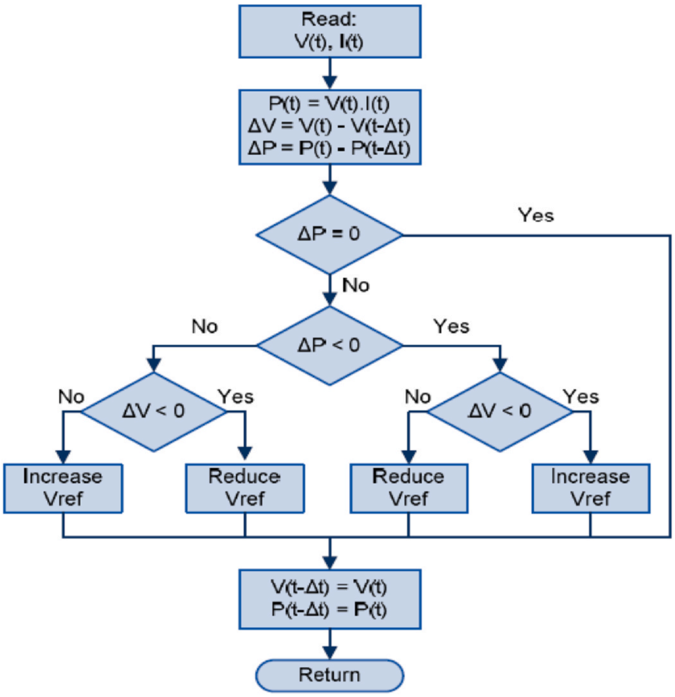

Fig. 6. Flowchart for the P&O algorithm [32].

adjust the reference values of current and voltage, respectively. In this way, the reference values are adjusted each time to allow the consistent supply of AC power provided with an ample supply of inputs. The modeling of the VSC main controller is shown in Fig. 8.

3.4. PV array

The modeling of the PV array module is demonstrated in Fig. 9, where 2 inputs are the core needs to make a PV cell in an active mode. The temperature and solar irradiance are put in the PV cell with the help of a function generator, signal builder, or by external Excel sheets. The output is generated in terms of voltages and current.

The PV module used in this simulation is SunPower SPR-305E-WHT-D, whose plate specification values are given in Table 3. Even the user-defined values can be used in the PV array module if a specific company is not given in the list of MATLAB PV array models. However, for the existing model, the array data consists of 66 parallel strings and 5 series-connected modules per string

4. Results and discussion

4.1. Characteristics profile of direct normal irradiance

The characteristic profiles of DNI for maximum, minimum, and full-day (24-h) conditions are shown in Figs. 10–12, respectively. The

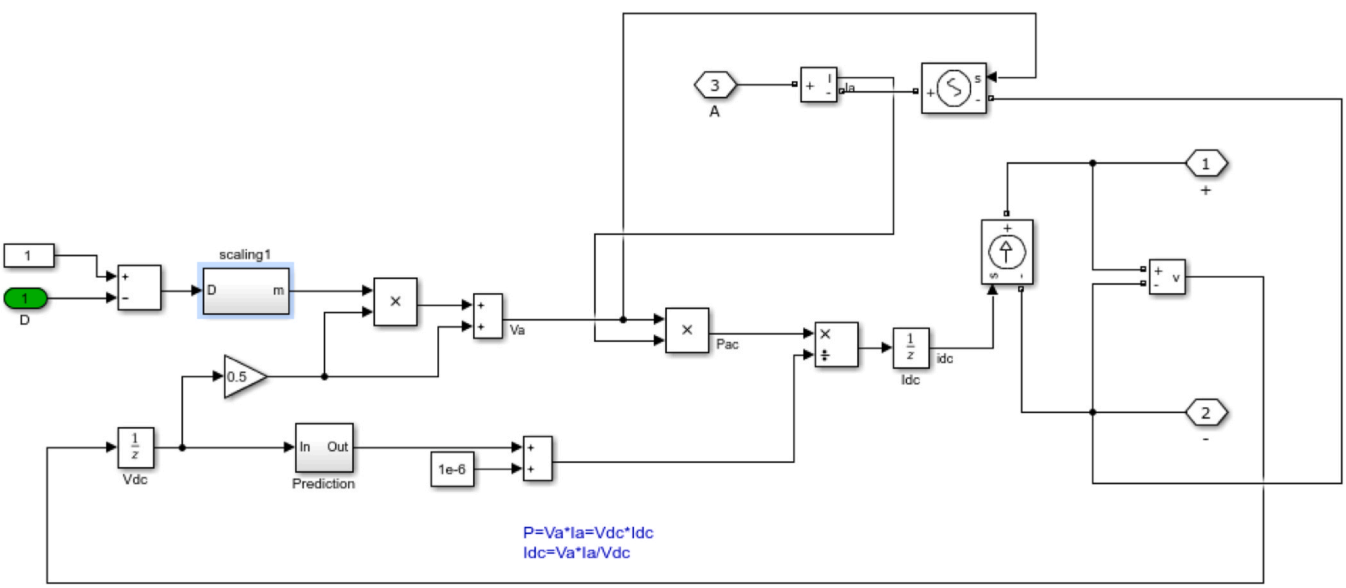

Fig. 7. Modeling of a boost converter.

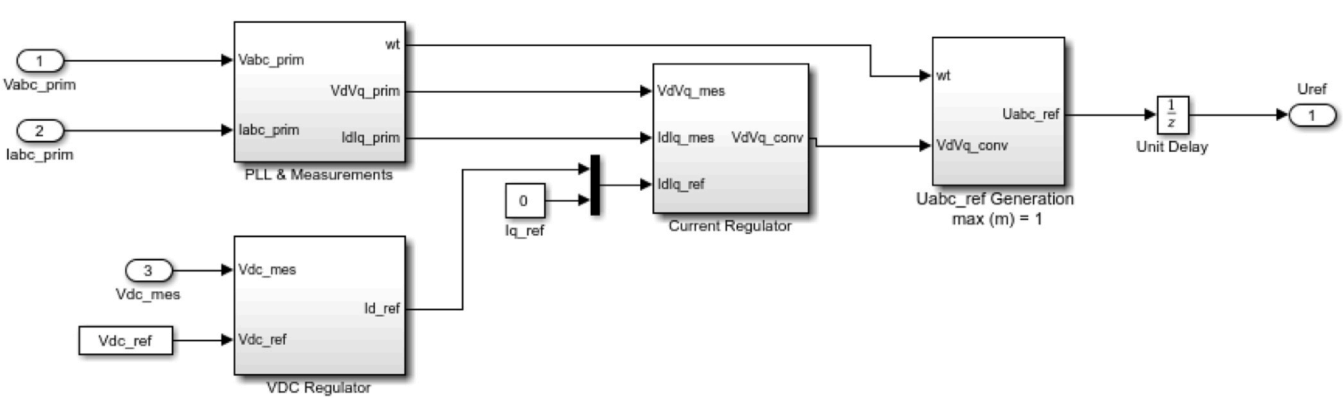

Fig. 8. Modeling of a VSC main controller. VSC = voltage source converter.

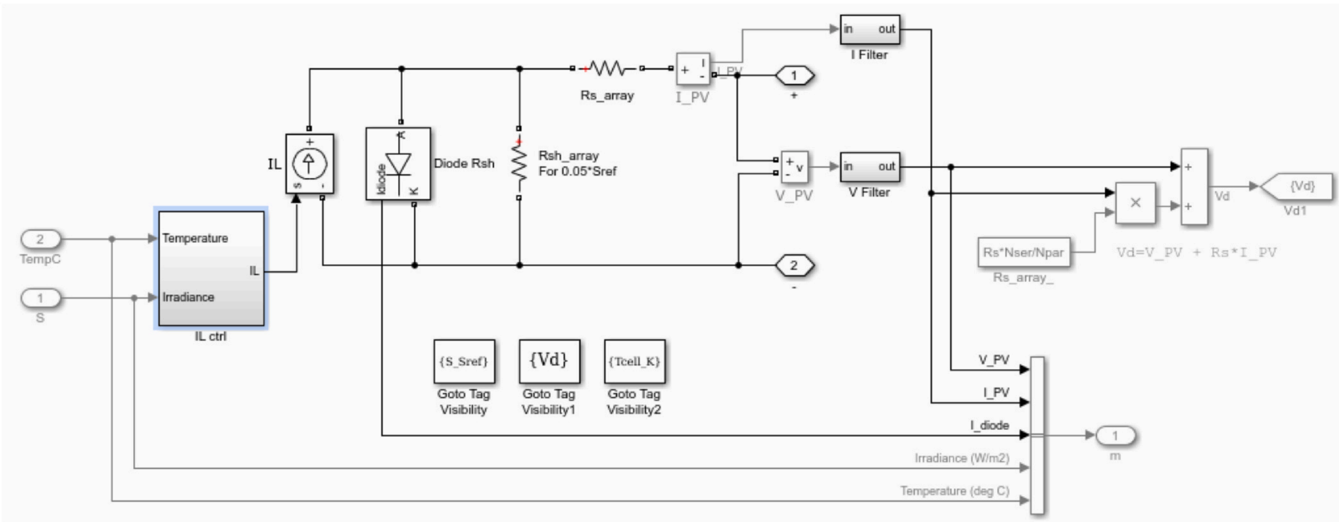

Fig. 9. PV array modeling. PV = photovoltaic.

Table 3
Solar plate specification parameters

Plate specification of SunPower SPR-305E-WHT-D	Values
Maximum power (W)	305.226
Open circuit voltage Voc (V)	64.2
Voltage at maximum power point Vmp (V)	54.7
Temperature coefficient of Voc (%/deg.C)	0.27269
Cell per module (Ncell)	96
Short circuit current Isc (A)	5.96
Current at maximum power point Imp (A)	5.98
Temperature coefficient of Isc (%/deg.C)	0.061745
Light-generated current IL (A)	6.0092
Diode saturation current I0 (A)	6.3014*10ˆ-12
Diode ideality factor n	0.94504
Shunt resistance Rsh (ohm)	269.5934
Series resistance (ohms)	0.37152

characteristics profile includes the solar irradiance, PV module surface temperature, average power, voltage profile, and the duty cycle.

4.2. Effect of direct normal irradiance

The data obtained from the Power Data Access Viewer of NASA is DNR, which is converted to DNI. The power generated is shown in Fig. 13 for maximum DNI sunshine, minimum DNI sunshine, and 24-h DNI, respectively.

The characteristics profile of DI is given in Figs. 14–16 for maximum, minimum, and average sunshine.

4.3. Effect of diffused irradiance

The data fetched from the Power Data Access Viewer of NASA is the DR, which is the radiation striking on a solar panel with less than 10%

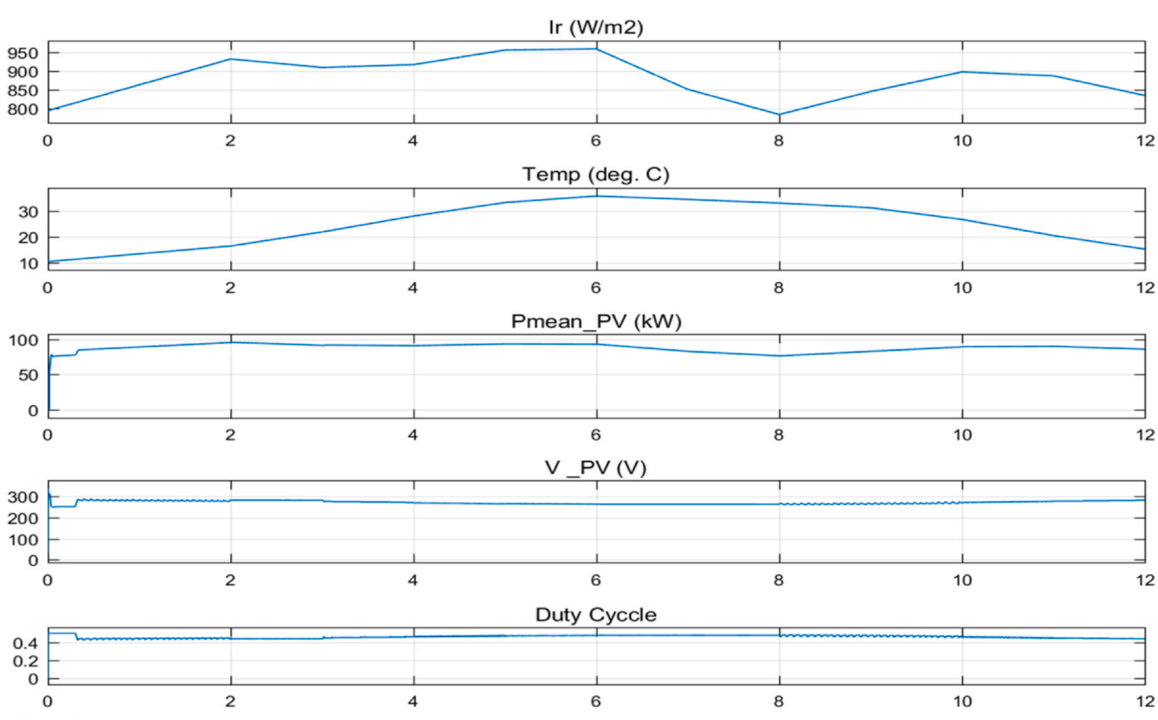

Fig. 10. Characteristics profile of DNI for maximum sunshine. DNI = direct normal irradiance.

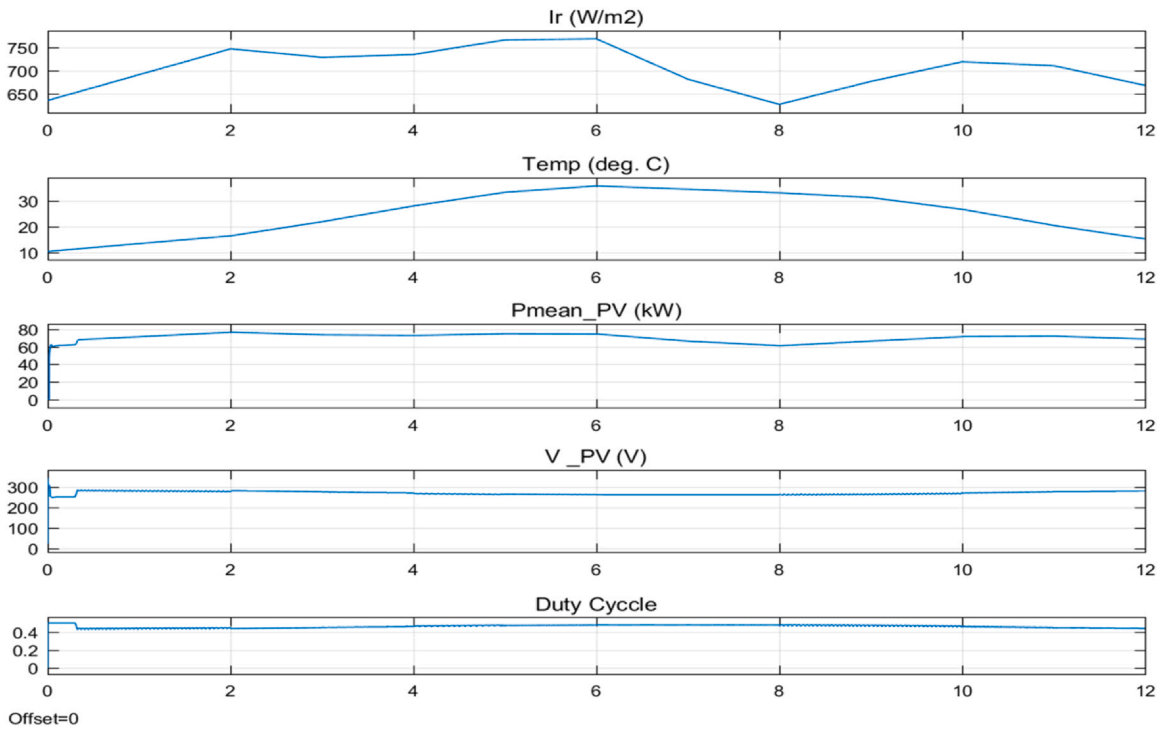

Fig. 11. Characteristics profile of DNI for minimum sunshine. DNI = direct normal irradiance.

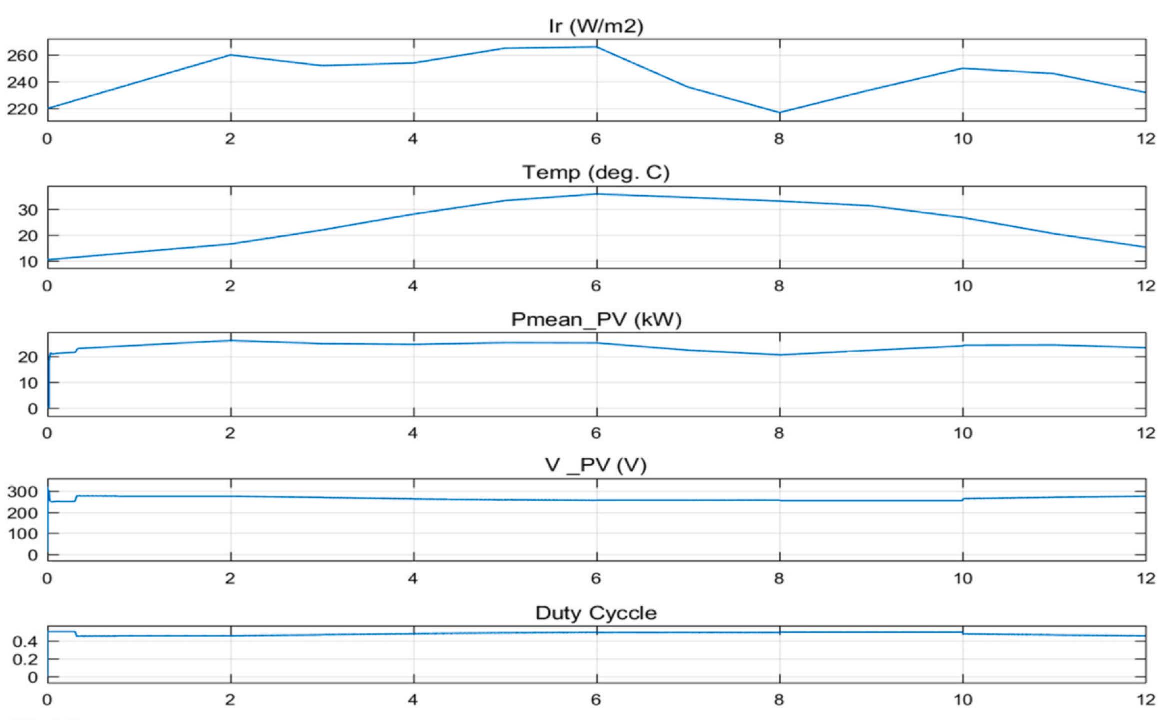

Fig. 12. Characteristics profile of DNI for full day (24 h). DNI = direct normal irradiance.

cloud shatter, regardless of DNR that strike the panel perpendicularly. Therefore, the DR is converted to DI for the correct input parameter of the PV solar model of a 100-kW. The power generated curves supplied to a grid are represented in Fig. 17, respectively, to maximum sunshine DI, minimum sunshine DI, and average sunshine DI.

4.4. PV module temperature modeling

The operating temperature of a PV module is a critical factor influencing its electrical performance and overall energy yield. In practical conditions, the module temperature is not equal to the ambient

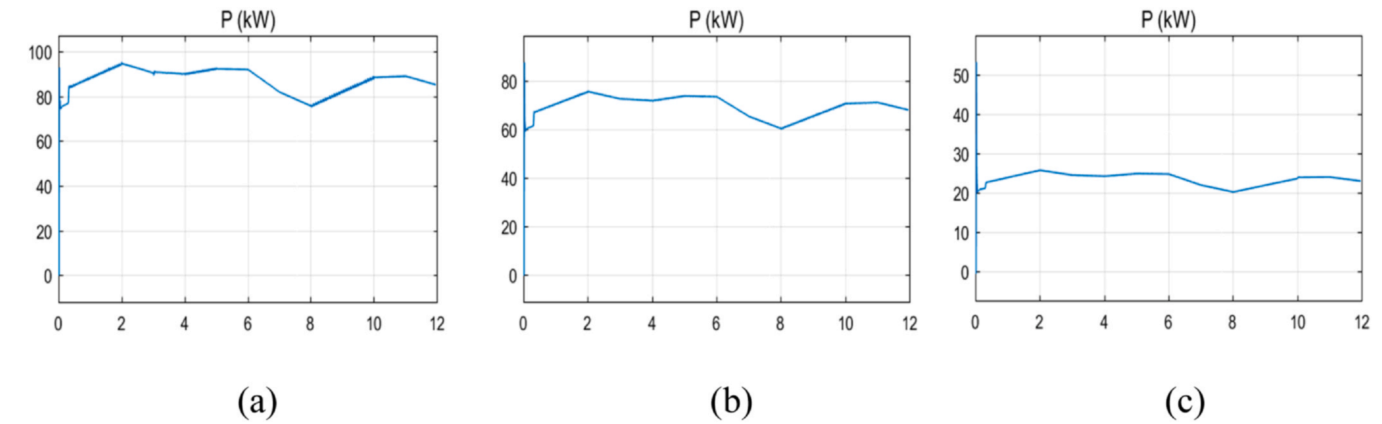

Fig. 13. Power generated to the grid via bus (a)maximum sunshine DNI, (b) minimum sunshine DNI, (c) full day (24 h) DNI. DNI = direct normal irradiance.

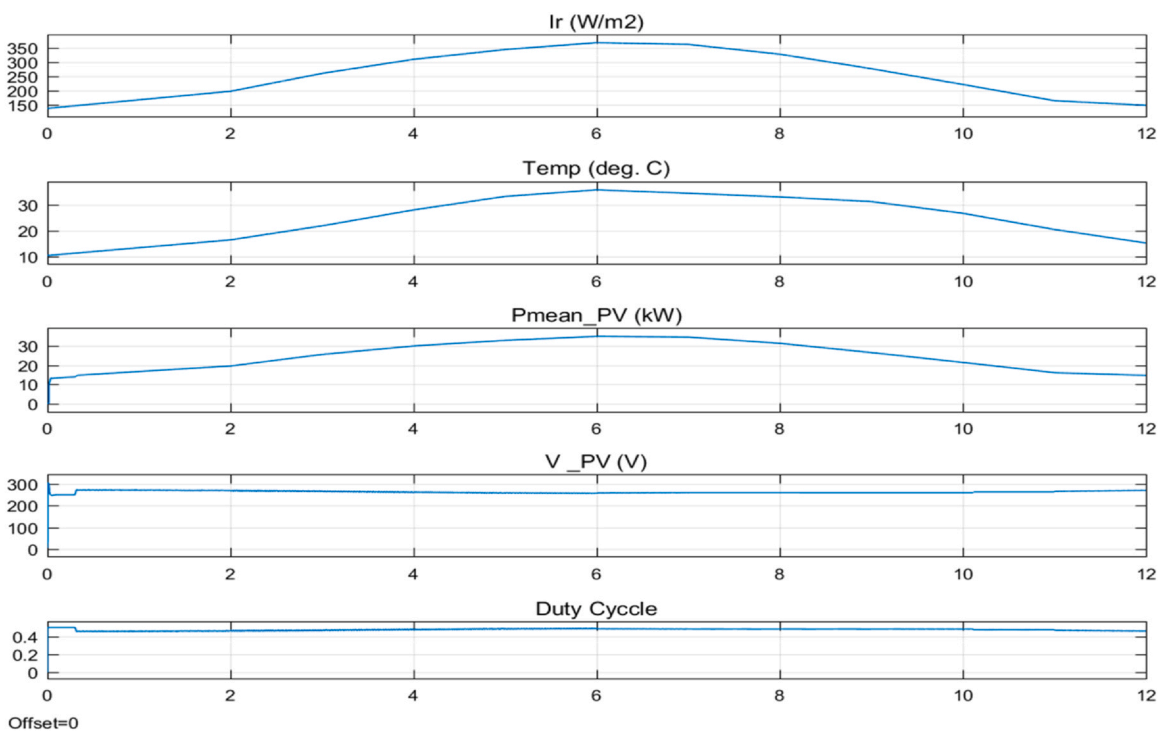

Fig. 14. Characteristics profile of DI for maximum sunshine. DI = diffused irradiance.

temperature, but is governed by a combination of environmental and system-specific parameters, including solar irradiance, ambient temperature, wind speed, and mounting configuration.

In this study, long-term climatological temperature data (35-year averages) obtained from the NASA POWER Data Access Viewer are used as the primary thermal input. These values represent ambient conditions for the selected location (Bahawalpur region) and provide a stable basis for comparative simulation across different irradiance scenarios, as shown in Fig. 18.

To account for the physical relationship between ambient conditions and PV module temperature, a standard Nominal Operating Cell Temperature (NOCT)-based formulation is considered. The module temperature can be expressed as:

$$T_{cell} = T_{ambient} + \frac{NOCT - 20}{80} \times G\# \tag{1}$$

where T_{cell} is the PV cell temperature (°C), $T_{ambient}$is the ambient temperature (°C), G is the incident solar irradiance (W/m^2), and NOCT represents the nominal operating cell temperature under standard test conditions. However, due to the use of monthly averaged long-term data and the absence of high-resolution inputs such as wind speed and installation-specific parameters, the above model is not fully implemented in the simulation. Instead, ambient temperature is used as a baseline approximation for PV module temperature. This simplification enables consistent comparison across multiple irradiance cases while maintaining computational efficiency. It is important to note that this assumption may introduce deviations from actual operating conditions, particularly under high irradiance or low wind scenarios where module temperature typically exceeds ambient temperature. Therefore, the temperature-related results presented in this study should be interpreted as indicative trends rather than exact field values.

4.5. Effect of temperature

It is a common mistake among renewable contractors and investors that high temperature will lead to high power rating results; however,

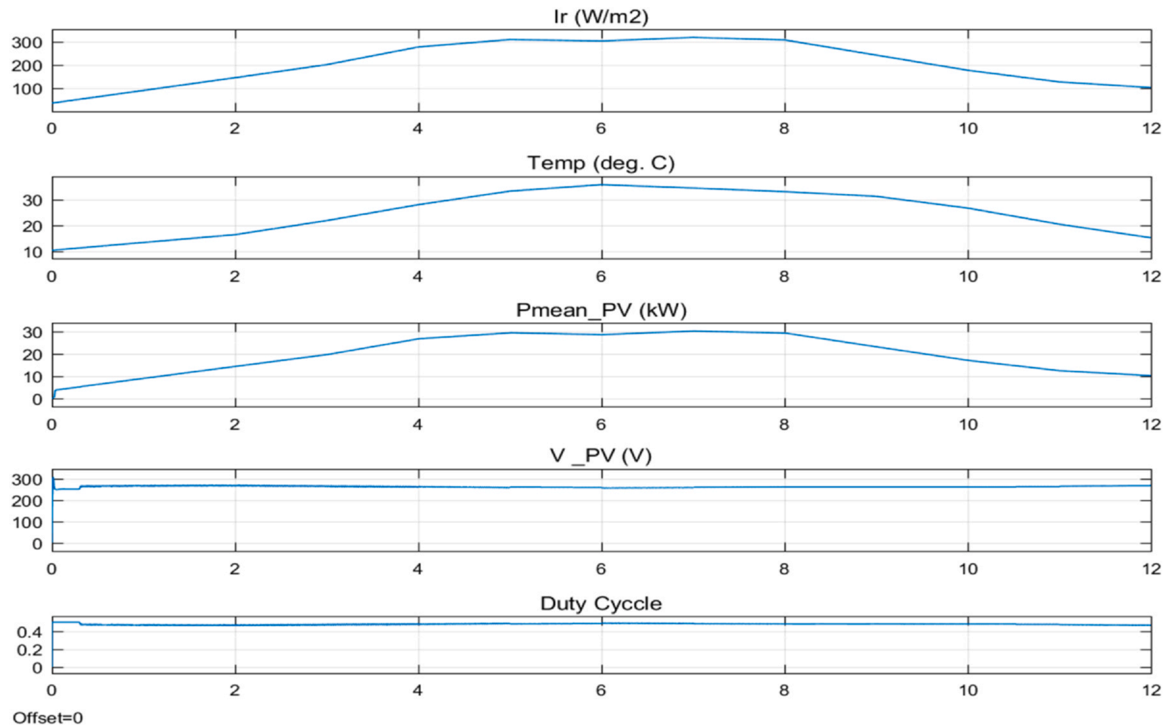

Fig. 15. Characteristics profile of DI for minimum sunshine. DI = diffused irradiance.

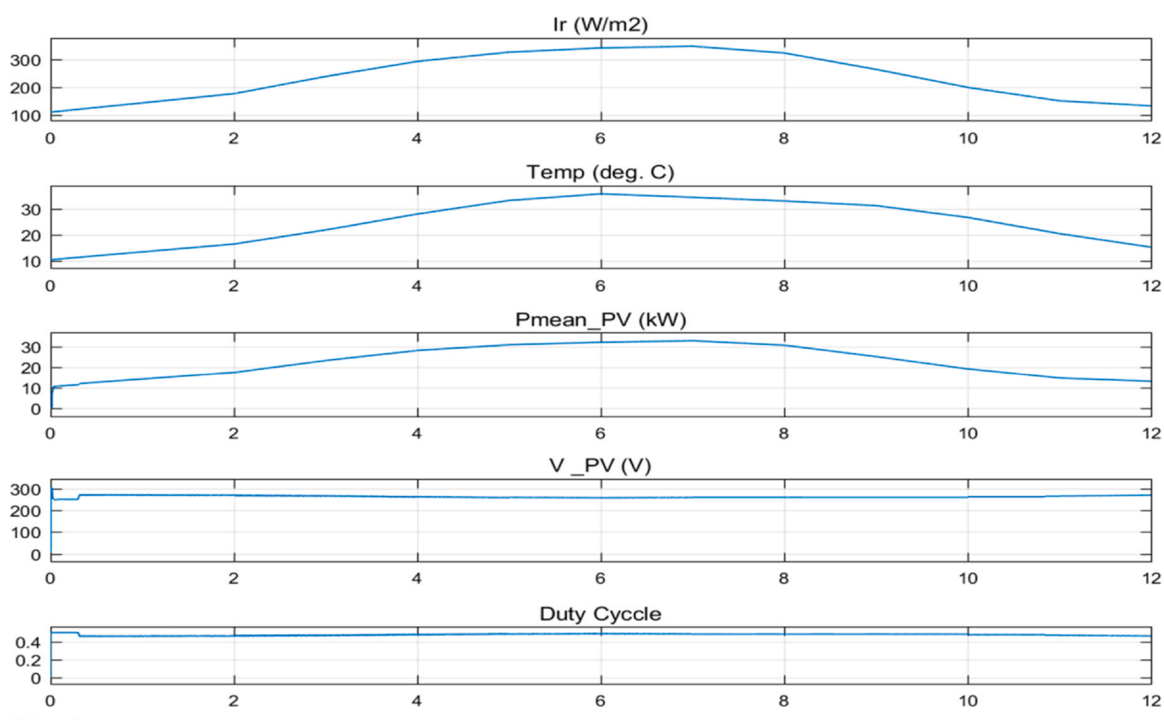

Fig. 16. Characteristics profile of DI for average sunshine. DI = diffused irradiance.

the case is not the same in the theory of solar PV panels. This is elaborated in the Fig. 19, the maximum and minimum temperature ranges. It is to be noted that the minimum DNI is used as the other input, combined with the initial input of temperature to the PV solar array, for the clear judgment of results. Here, DNI minimum values are treated as the constant parameter.

A significant amount of work has been conducted on improving the energy excavation form the PV panels by improving the abilities of inverters and MPPT techniques. This may count to only registering the utilization of generated energy with maximum efficiencies of onto 99%, at most [33]. Also, new techniques and algorithms are being introduced for improving the efficiencies beyond 99% [34]. Similar

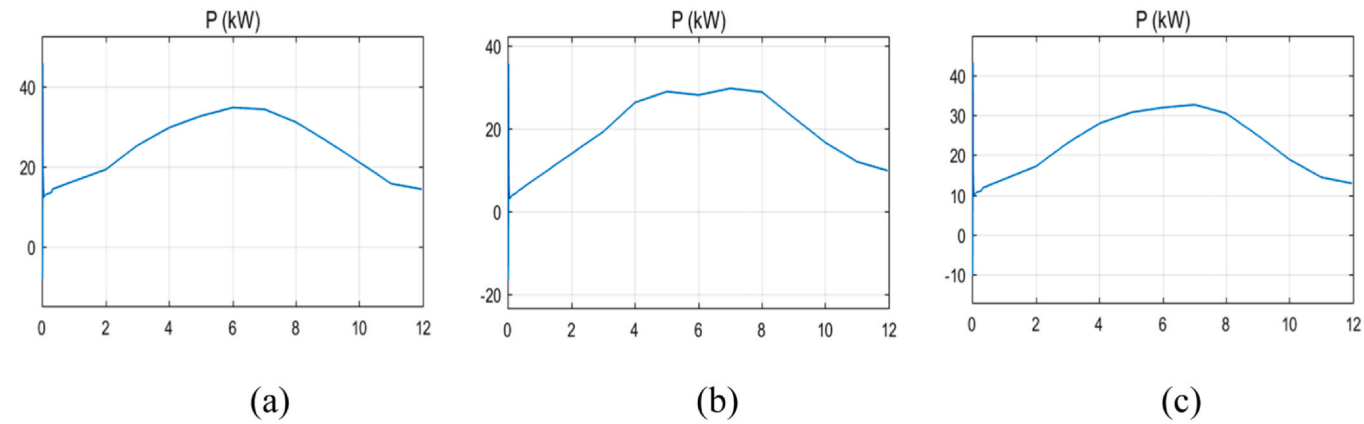

Fig. 17. Power generated to the grid via bus (a) maximum sunshine DI, (b) minimum sunshine DNI, (c) full day (24 h) DNI. DI = diffused irradiance; DNI = direct normal irradiance.

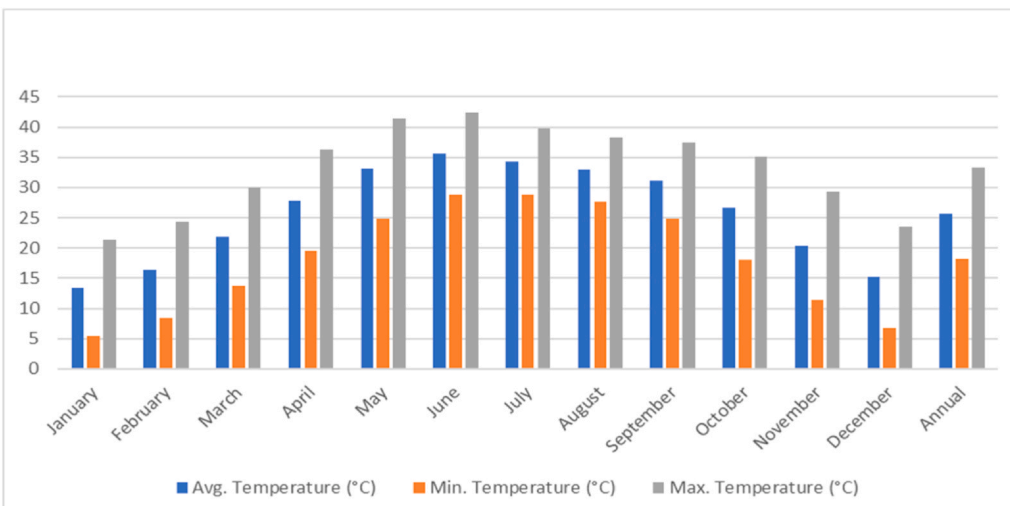

Fig. 18. Climatology surface temperature ranges of a PV module for Bahawalpur City. PV = photovoltaic.

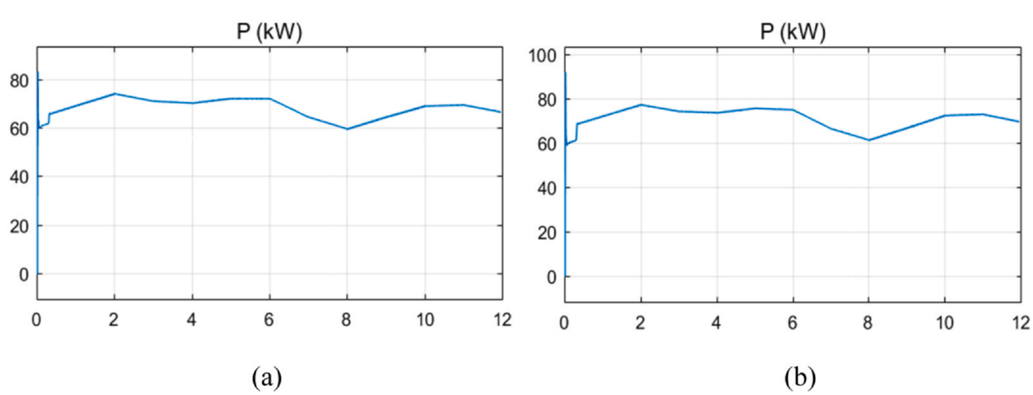

Fig. 19. Power generated to the grid network via bus (a), maximum temperature (b).

trends in DNI variation have been reported in previous studies [35,36], where the diurnal and seasonal fluctuations significantly influence solar energy generation. These works further emphasize the importance of accurate irradiance profiling for reliable performance assessment of PV systems. However, the need should be to validate and improve the energy production of solar energy for most. Therefore, to realize the fact, the results indicate finding the true power generated or lost due to environmental factors, infrastructure, and material properties, for any location in the world, in real-time. Once it is realized, surely a large amount of work will switch to revitalize and increase the production of energy from the existing to new installations of PV solar power plants.

Comparison of the predicted annual yield with the measured plant output is presented in Table 4. The results reveal a significant

Table 4
Predicted yield 100 MW simulation reframed for 100 MW versus measured plant output year 2025

Month	Power output (kW) full day (24 hr)		Monthly energy yield (kWh)		Monthly energy yield (MWh)
	Direct normal irradiance	Diffused irradiance	Direct normal irradiance	Diffused irradiance	Plane of array irradiance
Jan	22.2	12.5	16516.8	9300	12227.06
Feb	25.3	18	18823.2	13392	12043.39
Mar	23.7	21.7	17632.8	16144.8	16017.45
Apr	24.1	28.8	17930.4	21427.2	15246.55
May	24.9	30.6	18525.6	22766.4	14732.46
Jun	24.8	31.1	18451.2	23138.4	13052.29
Jul	21.3	31.2	15847.2	23212.8	13439.48
Aug	20	30	14880	22320	12646.89
Sep	23.1	24.6	17186.4	18302.4	13497.01
Oct	23.9	18.3	17781.6	13615.2	13905.11
Nov	23.5	14.5	17484	10788	9581.83
Dec	23.3	12.9	17335.2	9597.6	10619.95
Annual energy yield (kWh) for 100 kW			*208394.4*	*204004.8*	157009.47
Annual energy yield (MWh) for 100 kW			*208.3944*	*204.0048*	
Annual energy yield (MWh) for 100 MW			208394.4	204004.8	

Table 5
Monthly energy yield estimation for the data set

Reframed contributions for a yield-estimation method				
100 kW grid-connected solar PV system simulation				
Month	Power output (kW) full day (FD) (24 hr)		Monthly energy yield (kWh)	
	Direct normal irradiance	Diffused irradiance	Direct normal irradiance (FD)	Diffused irradiance (FD)
Jan	22.2	12.5	688.2	387.5
Feb	25.3	18	784.3	558
Mar	23.7	21.7	734.7	672.7
Apr	24.1	28.8	747.1	892.8
May	24.9	30.6	771.9	948.6
Jun	24.8	31.1	768.8	964.1
Jul	21.3	31.2	660.3	967.2
Aug	20	30	620	930
Sep	23.1	24.6	716.1	762.6
Oct	23.9	18.3	740.9	567.3
Nov	23.5	14.5	728.5	449.5
Dec	23.3	12.9	722.3	399.9
Annual energy yield (kWh)			8683.1	8500.2

performance gap between the installed capacity (100 MW) and the operational output (18 MW) of the Quaid-e-Azam Solar Park. In 2025, the measured annual energy yield is approximately 18% of the predicted value obtained from the 100 MW simulation. This also supports the accuracy of the dataset, where 35 years of NASA climatology data are condensed into 12 monthly representative points. The dataset was sourced from the NASA Power Data Archive under official ticket #97881, reflecting the data constraints considered in this study.

Table 5 further provides month-wise predicted energy yields from the simulation, separated into DNI and DI, along with measured plant output for validation under real operating conditions. The comparison emphasizes the discrepancy between expected and actual generation in large-scale PV systems. Under optimal conditions, DNI can contribute up to 87.07% of daily maximum power, with a corresponding loss of 12.93% in a 100 kW PV system.

The comparison analysis for the results of output power for the variables of radiation may be understood form Fig. 20, where the 6 parameters of irradiance are illustrated in the graph. It is evident that the maximum power achievable is close to 90%. Also, the 3 temperature parameters are shown in Fig. 21 for the output power to the grid. It highlights that maximum temperature may reduce power efficiency due to optimal plate operating temperatures, which may not satisfy the high temperature ranges.

These results highlight the importance of engineering optimization to adopt best engineering practices and ensure the accuracy of the solar tracking system to meet the percentages in the fields coherent to the simulation results. This can only be achieved in the real PV power plants to capture the maximum DNI that strikes the solar panels. Whereas, this purpose is achieved in only 15–25% to increase the production of energy by different methods and techniques. Table 5 complements this analysis by providing a statistical and descriptive summary of the simulated outputs, including key indicators such as mean, peak, and variability of power generation across different irradiance and temperature scenarios. Negative power values observed in initial simulations were associated with transient behavior during system initialization and controller settling. These values have been excluded by applying a steady-state filtering window, and the statistical results presented in Table 5 reflect steady operating conditions.

Generally, investors may be reluctant to show a prompt interest in the solar development business and its activities because of their limited understanding. Therefore, a large part of development in the sector of solar energy is implemented by the governments. Normally, investors may not achieve expected returns on solar due to least understandings of technical contracts. For instance, a investors allocate capital to erect a PV solar plant, but the return on energy is far lower due to initial high investments. Therefore, this research is useful for investors to understand the true solar potential for a specific area when signing heavy deals and contracts before ending up with low investment returns on their capital. The proposed framework can be utilized by researchers, engineers, and policymakers to evaluate solar PV feasibility for different locations based on long-term climatic data. This approach supports data-driven decision-making in the planning and assessment of PV solar power systems.

The performance analysis conducted for the PV solar power plant may be extended beyond the boundaries of this research. As one may only need to use the manufactured solar panel specification values of the installed company's plate and repeat the whole simulation by using the user-default inputs to their specific region. In this way, the performance of a solar power plant is measured from climatology-based simulation to the practical field results. Therefore, the validation of results may help to improve the energy production of existing PV solar power plants.

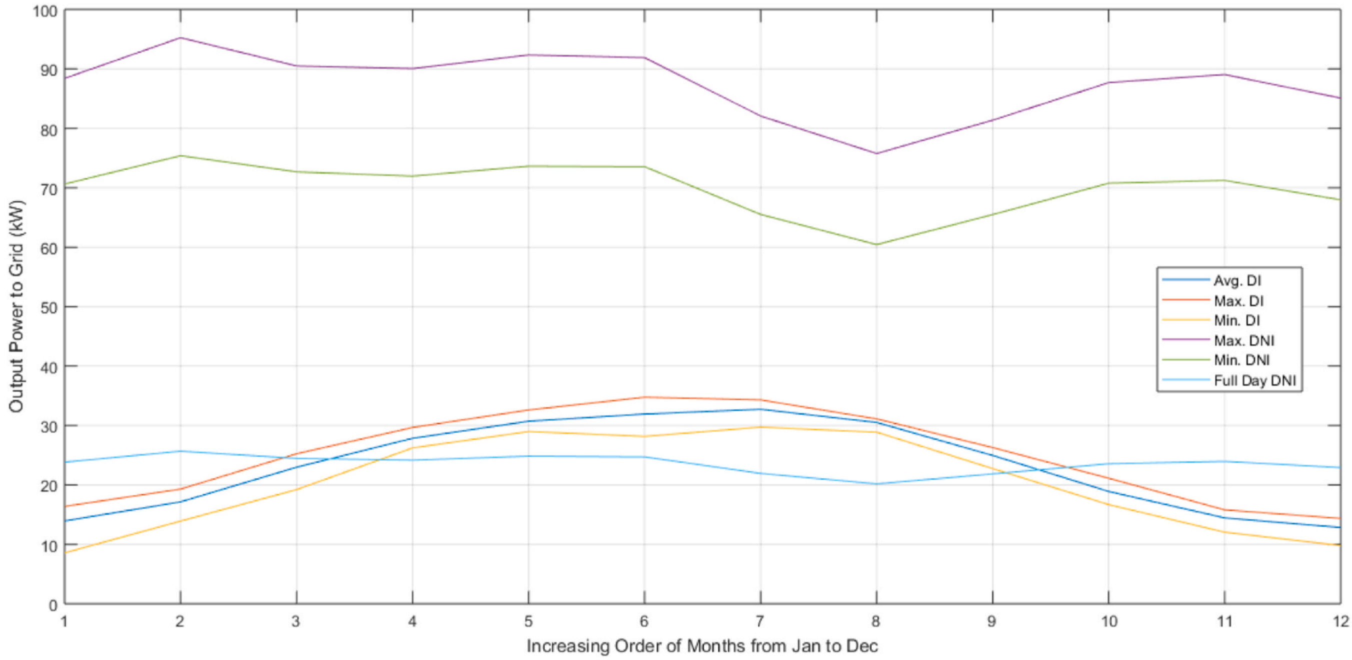

Fig. 20. Comparison analysis for the output power to the grid for 6 parameters of irradiance. DI = diffused irradiance; DNI = direct normal irradiance.

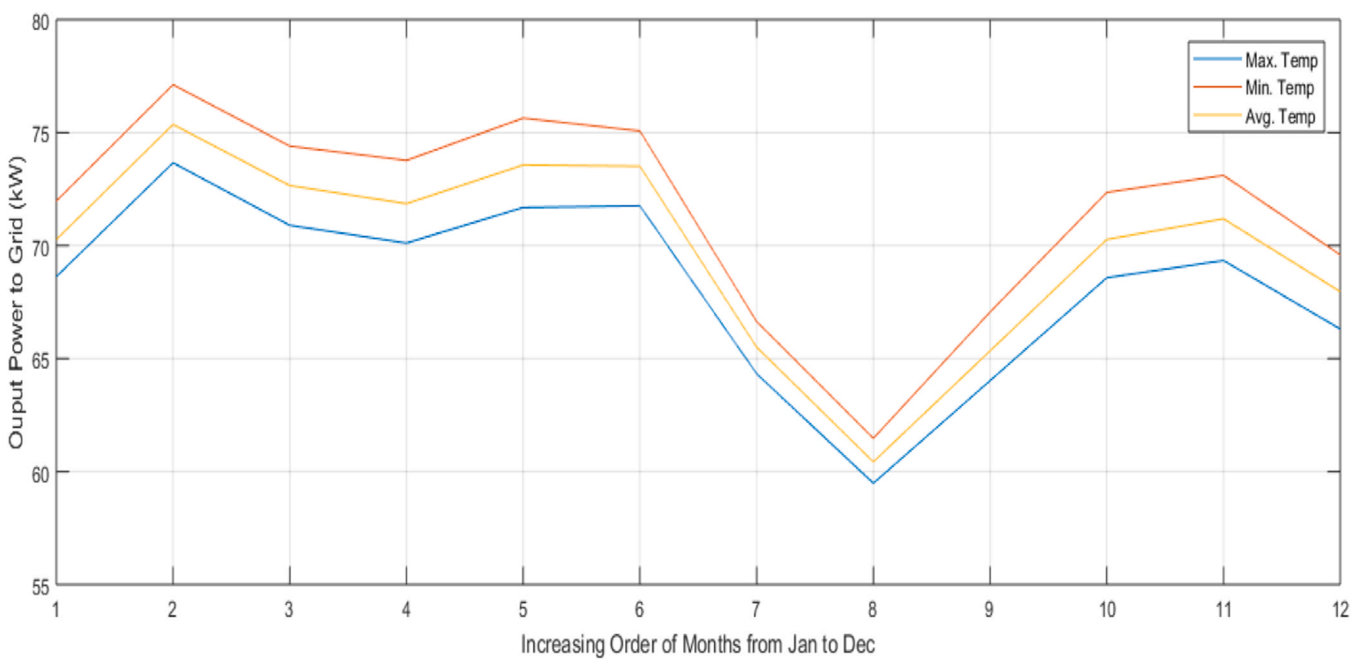

Fig. 21. Comparison analysis for the output power to the grid for 3 parameters of temperature.

5. Conclusion

The scheme introduced is to validate the difference in the power ratings from the installed to the run capacity of a PV solar power plant. It leads to the identification of the solar power potential of any locality around the world. Further, it develops to address the technical aspects relevant to the PV solar power plant, which proves the credibility of the research in climatology-based simulation. The MATLAB/SIMULINK model of a 100-kW grid-connected produces practical results based on NASA data set points. It benefited the analysis to statistical and graphical terms, which states about finding a true solar potential for the installation of a PV solar power plant. This may help to support the feasibility assessment of the feasibility studies without spending extravagant money on physical land surveys. The findings suggest the gap between simulated (expected) power output and the actual generation of a PV solar power plant. This difference represents the portion of potential energy that is not realized in practice due to combined effects of environmental variability, system losses, and non-ideal operating conditions. In standard PV system analysis, such discrepancies are commonly associated with loss mechanisms and performance ratios. For the Bahawalpur case study, the results demonstrate a clear difference between theoretical generation potential and observed plant performance. This research estimates the theoretical generation potential that a PV solar power plant may convert electrical power from solar energy. This concept can be further investigated in future studies by analyzing the differences between simulation and practical results using an existing 100 kW grid-connected PV model for next-generation energy applications.

This study uses 35-year climatological data aggregated into monthly averages, which do not capture short-term variability or year-to-year fluctuations. Several practical factors are not explicitly modeled, including system orientation, soiling, shading, degradation, curtailment, and detailed thermal effects. In addition, satellite-derived data may introduce uncertainties compared to ground measurements, and results are not cross-validated with other simulation tools. Therefore, the proposed method should be interpreted as a preliminary feasibility and yield estimation approach, rather than a detailed performance analysis.

CRediT authorship contribution statement

Md Shafiullah: Writing – review & editing, Visualization, Investigation, Formal analysis. **Muhammad Shahid Mastoi:** Visualization, Formal analysis, Data curation. **Azfar Rasool:** Writing – original draft, Validation, Software, Methodology, Data curation, Conceptualization. **Muhammad Amjad:** Writing – review & editing, Visualization, Supervision, Resources, Project administration. **Mannan Hassan:** Writing – review & editing, Visualization, Validation, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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